MIL-S-49181(CR)
18 April 1979

MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT
S-583( )/TRC-151(V)

This specification is approved for use by Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of the Radio Terminal Set AN/TRC-151(V) which is capable of providing a dual multiplex terminal for 12 or 24 channels of voice frequency communications for radio or cable transmission. (See 6.1 and 6.5)

1.2 Description.- Shelter, Electrical Equipment S-583( )/TRC-151 (V), herinafter called the shelter facility, includes a modified S-280() /G shelter and all racks, brackets, shelves, tie-downs, patch panel, electrical components and wiring to enable installation and operation of the assemblage components of the AN/TRC-151(V).

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications, Research and Development Command, ATTN: DRDCO-CM-DM Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

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SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging Methods of |
| MIL-C-3885 | Cable Assemblies and Cord Assemblies Electrical (for use in Electronic and Associated Electrical Equipment) |
| MIL-W-6858 | Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickle Alloys, Heat Resisting Alloys & Titanium, Spot & Seam |
| MIL-T-7928 | Terminals, Lug & Splice, Crimp Style Copper |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-W-45205 | Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor |
| MIL-S-55507 | Shelter, Electrical Equipment (with or without equipment), Packaging of |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures & Tables for Inspection by attributes. |
| MIL-STD-252 | Wire Equipment, Classification of Visual & Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-1235 | Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes |
| MIL-STD-1474(MI) | Noise Limits for Army Material |

DRAWINGS

MATERIAL READINESS COMMAND

| | |
|---|---|
| DL-SC-B-913777 | Shelter, Electrical Equipment S-583( )/ TRC-151(V) |
| SC-D-913818 | Power Schematic |
| SC-F-913819 | Signal Schematic |



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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies)

3. REQUIREMENTS

3.1 First article.- When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with the drawings as listed on DL-SC-B-913777 and the requirements specified herein. (See 3.25)

3.3 Electrical installation.- (See 4.6)

3.3.1 Power system wiring and components.- The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- Continuity shall exist between each pin of the power entrance connectors and respective circuit terminations and shall be less than 3 ohms. (See 4.6.1 and 4.6.1.1)

3.3.1.2 Dielectric breakdown voltage.- The insulation of all power wires, cables, and components shall withstand, without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than 30 seconds. (See 4.6.1 and 4.6.1.2)

3.3.1.3 Insulation resistance.- The resistance of insulation between any two circuit conductors and also between each of the four ungrounded power line conductors and shelter facility ground terminal shall not be less than 100 meghoms. (See 4.6.1 and 4.6.1.3)

3.3.2 Power distribution and operational functions.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as shown in the power schematic drawing when tested in accordance with 4.6.2.

3.3.3 Signal (audio) system wiring and components.- The completed in-place signal system installation shall meet the following requirements:

3.3.3.1 Continuity.- Continuity shall exist between end terminations of all signal wires and shall be less than 5 ohms. (See 4.6.3 and 4.6.3.1)

3.3.3.2 Dielectric strength of insulation.- The insulation of all signal wires, cables, and components shall withstand, without flashover or



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insulation breakdown, a potential of 500 volts DC applied continuously for not less than 5 seconds. (See 4.6.3.2)

3.3.3.3 Insulation resistance.- The resistance of insulation of each signal line conductor and all other signal line conductors connected to ground shall not be less than 100 megohms. (See 4.6.3.3)

3.3.4 Video (rf) system wiring and components.- Installation of video and rf cable assemblies and connecting components (including plugs, connectors, adapters, fittings, receptacles and lightning protectors) shall be in accordance with the drawings. The completed in-place video and rf system installation shall meet the following requirements:

3.3.4.1 Continuity.- The point-to-point continuity of all rf cable assemblies shall not exceed 1 ohm. (See 4.6.4 and 4.6.4.1)

3.3.4.2 Dielectric and insulation resistance.- The insulation of all video and rf cables, components and lightning protectors shall withstand the following potentials with the insulation resistance not less than 100 megohms (See 4.6.4.2):

a. 1500 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements removed.

b. 1100 volts DC applied continuously for not less than 30 seconds to all video cable assemblies terminated at the video entrance box with lightning protector elements in place.

c. 1700 volts RMS, 25 - 100 cycles, applied continuously for not less than 30 seconds to all rf transmission lines terminated at the antenna entrance box.

d. 1500 volts DC applied continuously for not less than 5 seconds to all video cable assemblies terminated at the video and radio patch panel.

3.3.5 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull-out requirements of MIL-C-3885. (See 4.4)

3.3.6 Suppression of electromagnetic interference.- The motors in the ventilating equipment used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4)



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3.4 Systems operation.- With all operational equipment of the AN/TRC-151(V) assemblage installed, the operational requirements of all power and signal cable assemblies, switches, connectors, and operational equipment shall be met when tested in accordance with 4.7.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicle without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.8)

3.6 Railroad transportation.- The shelter facility shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail and shall be subjected to and withstand the tests of 4.9 without sustaining any permanent damage such as buckling, cracking, delamination or other permanent deformation.

3.7 Drop tests.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.10)

3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.11. During and after the test there shall be no leakage into the facility or its walls, floor, roof or doors.

3.9 Fording.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the facility or its walls, floor or doors.

3.10 Humidity.- Hardware sample shall have no evidence of corrosion rust or deterioration after testing. (See 4.13)

3.11 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and, when installed each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.14. There shall be no damage or permanent distortion to the shelter mounting surface or insert head or threads and the inserts shall not have rotated from its original position. Insert threads shall be free of paint.

3.12 Acoustical Noise.- The acoustical noise shall not exceed the criteria specified in MIL-STD-1474A(MI), Table 2, Category F. (See 4.15)

3.13 Cleaning.- (See 3.25)

3.13.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable por-



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tions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.13.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware and other foreign material.

3.14 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 3.25 and 4.4).

3.15 Marking.- (See 3.25 and 4.4)

3.15.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.15.2 Nameplates.- Shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.15.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The location of the centers-of-gravity shall be determined by the contractor using the first article of production.

3.15.4 Interior and exterior marking.- All interior and exterior markings shall be as defined on the drawings.

3.16 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of MIL-STD-454. (See 3.25 and 4.4)

3.17 Seal, water vapor.- All joints of the framing for the exhaust blowers, power, signal and antenna entrance boxes and air conditioner ports shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws and fast-



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shipment and storage shall consist of the examinations specified in Table VII.

ners shall be dipped in sealer prior to installation. The sealer shall be a one-part sealant, Silastic 738 as made by Dow Corning, or equal using the manufacture recommended primer. (See 3.25 and 4.18)

3.18 Securing of parts. Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.25 and 4.18)

3.19 Soldering. (See 3.25 and 4.18)

3.19.1 Solder and flux. Solder and flux shall be as defined on the drawings.

3.19.2 Acid or acid salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception permitted for preliminary tinning of electrical connections and for soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.19.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign materials remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.20 Tropicalization of material. Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings as follows:

3.20.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.20.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrica-



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tion, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.20.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.20.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.7)

3.21 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-45205 CL.B as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.22 Wiring and cabling.- (See 3.25 and 4..8) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.22.1 Slack.- Wires and cables shall be as short as practicable except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected or disconnected.

c. To facilitate field repair of broken or cut wires.

d. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.22.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires



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shall not be bent sharply where they enter insulation material.

3.22.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.22.4 Solderless terminal lug connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of MIL-T-7928. (See 4.4 and 4.16)

3.22.5 Identification of wiring.- Wiring shall be identified by color coding and wire pair marking, as indicated on the drawings.

3.22.6 Cord Assemblies.- Power cable and cord assemblies shall be fabricated in accordance with the drawings and meet the requirements and tests of MIL-C-3885. (See 4.4)

3.22.7 Testing.- Power and signal wiring of the S-583( )/TRC-151(V) shall be subjected to and pass the tests described in 4.6.

3.23 Suppression of radio frequency interference.- The motors in the exhaust fans, fluorescent lighting and other electrical devices used in the shelter facility shall meet the emanation limits of MIL-STD-461 for tactical support equipment, if other than those items defined by the drawings. (See 4.4)

3.24 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.17) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.25 Workmanship.- (See 4.18) The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

| | |
|---|---|
| 3.2 | Construction |
| 3.3 | Electrical installation |
| 3.11 | Mounting inserts |
| 3.13 | Cleaning |
| 3.14 | Finish |
| 3.15 | Marking |
| 3.16 | Riveting |
| 3.17 | Seal, water vapor |



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3.18 Securing of parts
3.19 Soldering
3.20 Tropicalization
3.21 Welding
3.22 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspections shall be classified as follows:

a. First article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.20)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First article units.- The contractor shall furnish one (1) first article unit of the complete S-583( )/TRC-151(V).

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.



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TABLE I - First Article Inspection

| Inspections | Rqr Para | Test Para |
|---|---|---|
| 1. Inspection covered by subsidiary documents | - | 4.4 |
| 2. Interchangeability | 3.24 | 4.17 |
| 3. Mounting Insert | 3.11 | 4.14.1 |
| 4. Solderless-terminal-lug-connection | 3.22.4 | 4.16 |
| 5. Humidity | 3.10 | 4.13 |
| 6. Visual & Mechanical | 3.25 | 4.18 |
| 7. Electrical Installation | 3.3 | 4.6 |
| 8. System Operation | 3.4 | 4.7 |
| 9. Acoustical Noise | 3.12 | 4.15 |
| 10. Vehicular transportation | 3.5 | 4.8 |
| 11. Railroad transportation | 3.6 | 4.9 |
| 12. Drop tests | 3.7 | 4.10 |
| 13. Watertightness | 3.8 | 4.11 |
| 14. Fording | 3.9 | 4.12 |
| 15. Repeat 7 - Electrical Installation | 3.3 | 4.6 |
| 16. Repeat 8 - System operation | 3.4 | 4.7 |

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

| Item | Where required |
|---|---|
| Cable & Cord Assemblies | 3.3.5 3.22.6 |
| Inteferance Suppression | 3.3.6 |
| Finish | 3.14 |
| Marking | 3.15 |
| Riveting | 3.16 |
| Welding | 3.21 |
| Solderless terminal lug connections | 3.22.4 |

4.5 Quality conformance inspection of equipment before packaging.- The contractor shall perform the inspection specified in paragraph 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsi-



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bility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A.- Group A inspection, shall be performed in the order listed in Table II.

TABLE II Group A Inspection

| Inspection | Reqt Para | Test Para | AQL % Major | AQL % Minor |
|---|---|---|---|---|
| Visual & Mechanical | 3.25 | 4.18 | 1.0 | 4.0 |
| Electrical Installation | 3.3 | 4.6 | 1.0 | * |

* All defects are major.

4.5.2 Group B Inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans.- The Group B sampling plans for the AQL's listed in Table III shall be as follows:

| AQL | Inspection Level |
|---|---|
| 4.0% | S-4 |
| 2.5% | S-4 |

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in the order listed in Table III.



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TABLE III - Group B Inspection

| Inspection | Reqt Para | Insp Para | AQL % (See 4.5.2.1) |
|---|---|---|---|
| Mounting Insert | 3.11 | 4.14.2 | 4.0 |
| Interchangeability | 3.24 | 4.17 | 4.0 |
| Watertightness | 3.8 | 4.11 | 2.5 |
| Fording | 3.9 | 4.12 | 2.5 |

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV.

TABLE IV - Group C Inspection

| Inspection | Reqt Para | Test Para |
|---|---|---|
| Solderless terminal lug connection | 3.22.4 | 4.16 |
| Mounting inserts | 3.11 | 4.14.1 |
| System operation | 3.4 | 4.7 |
| Acoustical noise | 3.12 | 4.15 |
| Vehicular transportation | 3.5 | 4.8 |
| Railroad transportation | 3.6 | 4.9 |
| Drop tests | 3.7 | 4.10 |
| Humidity | 3.10 | 4.13 |

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection with Group C for all inspections listed shall be as listed in Table IV. Units subject to Group C Drop Tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.11 and 4.12.

4.5.3.1 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a 1 foot length



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of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs to production.

4.5.3.2 Sampling for inspection of Drop Tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the truck and rail transportation tests of 4.5.3.3. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.3 Sampling for other inspection.- For inspection tests listed below one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the Government.

a. Mounting inserts

b. System operation

c. Acoustical noise

d. Vehicular transportation

e. Railroad transportation

f. Humidity

4.5.3.4 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.5.5 Alternate Group A Audit and Group B inspection procedures.- The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Sampling plans for the AQL's listed in Table II Group A inspection and Table III Group B inspection shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting



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production contractor shall notify the contracting officer in writing the method he intends to use. The method may not be changed during production without written approval of the Government.

4.6 Electrical installation tests.- The shelter facility shall be subjected to the tests of 4.6.1 through 4.6.4 below. Before applying power to the shelter facility, an electrical ground connection at the power entrance box shall be made. Power to be supplied shall be 115 VAC, 60 Hertz.

4.6.1 Power systems and components installation tests.- Refer to the Power Schematic Plate Drawing SC-D-913818.

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent and neon lamps.

(2) All fluorescent lamp starters.

(3) Blowers and heaters.

b. Remove power distribution box cover panel and disconnect conductors from the voltmeter terminals. Position or isolate the disconnected leads so as not to short to each other or facility ground.

4.6.1.1 Continuity of conductors.- There shall be point-to-point circuit continuity on all power line circuits and the resistance of the continuous circuit shall not exceed 3 ohms. Continuity resistance tests which meet this requirement shall be performed on each power line circuit in accordance with the following paragraphs. A calibrated low resistance ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position:

(1) All circuit breakers.

(2) Fluorescent lamp switches.

(3) Incandescent lamp switch.

(4) Blackout bypass switch (to bypass position).

(5) Blower switches.

b. Measure continuity resistance from pin 1 of 115V power connector J11 located at the power entrance box to each of the following:

(1) Brass (hot) contacts of all convenience, intercom, heater, power supply and blower power receptacles.



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(2) Black wire termination at each fluorescent and incandescent lamp.

(3) Pin 1 of 115V power connector OUT.

c. Place all controls listed in step a in the OFF position except main circuit breaker.

d. Repeat step b measuring continuity resistance from pin 2 of 115V power connector IN to the silver contacts of each power receptacle, white wire termination at each fluorescent and incandescent lamp, and pin 2 of 115V power connector OUT.

e. Place main circuit breaker in the OFF position.

f. Measure continuity resistance from pin 4 (ground) of 115V power connector IN to each of the following:

(1) Ground Terminal at each power receptacle.

(2) Ground Terminal located at power entrance box, power distribution box, equipment racks and raceways.

(3) Pin 4 of 115V power connector OUT.

4.6.1.2 Dielectric breakdown voltage.- The dielectric strength of insulations of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds. A non-destructive DC dielectric tester shall be used in performing the tests. Tests procedures shall be as follows:

CAUTION Take preventative measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test in order to eliminate the danger of electric shock.

a. Place all circuit controls listed under 4.6.1.1.a in the ON position except the fluorescent lamp switch.

b. Test between pin 1 and pin 4 of 115V power connector IN. Adjust tester to an output value of 500 volts DC for not less than 30 seconds. The dielectric strength requirements shall be met.

c. Place fluorescent lamp switch in the ON position.



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d. Repeat step b with tester connected first across pins 1 and 2 and then across pins 2 and 4 of 115V power connector IN.

4.6.1.3 Insulation resistance.- The insulation resistance between any two of three circuit conductors and also between each of the two ungrounded power line conductors and shelter facility ground shall be not less than 100 meghoms when tested in accordance with the following procedures. A meghometer or equivalent shall be used in performing the tests.

a. Place all circuit controls listed under 4.6.1.1.a in the ON position except the blackout bypass switch and fluorescent lamp switches.

b. Close shelter door securely.

c. Test between pins 1 and 4 of 115V power connector IN. The tester output voltage shall be between 100 and 500 volts DC. Maintain this potential until the meter indicator has ceased fluctuating and indicates a steady or increasing insulation resistance. The insulation resistance requirement shall be met.

d. Place the blackout bypass switch in the BYPASS position.

e. Repeat step c with the tester connected first across pins 1 and 2 and then across pins 2 and 4 of 115V power connector IN.

4.6.2 Power distributional and Operational function tests.- Power distributional and operational function tests shall be performed on each power line circuit in accordance with the following paragraphs.

4.6.2.1 Test preparation.- (Refer to power schematic plate)

a. Replace or reconnect all electrical loads and voltmeter which were removed or disconnected by 4.6.1.a and 4.6.1.b.

b. Close power distribution box panel cover.

c. Place all circuit breakers and switches in the OFF position.

d. Place an 115 volt AC test lamp with grounding plug into all convenience, heater, equipment and intercom receptacles.

e. Open outer exhaust port cover of blowers and insert AC line plug of blower into appropriate receptacle.

4.6.2.2 Test for 115 volt AC prime power.-



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(1) Quality conformance inspection of equipment before packaging.

a. Apply 115 volt AC 60 hertz, single phase prime power to the shelter facility at 115V power connector IN. The power indicator lamps at the power distribution panel shall light.

b. Measure line voltage by connecting a calibrated AC meter (0 - 150 volt scale) between pins 1 and 2 of 115 VAC power connector OUT.

c. Read voltage indicated on shelter facility voltmeter. The voltage read shall be within ± 2% of the voltage measure in step b above.

d. Place main circuit breaker in the ON position.

4.6.2.3 Tests of circuit breakers and switches.-

a. Lights circuit breaker (CB 1)

(1) Place circuit breaker CB 1 in the ON position. Neon lamp above CB 1 shall light.

(2) Place blackout bypass switch and fluorescent lamp switches in the ON position. All fluorescent lamps shall light.

(3) Place incandescent lamp switch in the ON position. All incandescent lamps shall light.

(4) Place CB 1 in the OFF position. All lamps mentioned in steps (1) thru (3) shall go out.

(5) Replace CB 1 in the ON position to restore light.

(6) With shelter door open, place blackout bypass switch in the BLACKOUT position. All fluorescent and incandescent lamps shall go out.

(7) Close shelter door. All fluorescent and incandescent lamps shall light.

(8) Place fluorescent lamp switches in the OFF position. All fluorescent lamps shall go out.

(9) Place incandescent lamp switch in the OFF position. All incandescent lamps shall go out.

(10) Open door and restore lights by placing blackout bypas switch, fluorescent light switch and incandescent light switch in the ON position.

b. Blowers circuit breaker (CB 2)



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(1) Place circuit breaker CB 2 in the ON position. Neon lamp above CB 2 shall light.

(2) Place roadside blower switch in the ON position. Roadside blower shall operate and shall exhaust air from the shelter.

(3) Place curbside blower switch in the ON position. Curbside blower shall operate and shall exhaust air from the shelter.

(4) Place circuit breaker CB 2 in the OFF position. The neon lamp above CB 2 shall go out and both blowers shall cease to operate.

c. Heater No. 1 circuit breaker (CB 3)

(1) Place circuit breaker CB 3 in the ON position. The neon lamp above tandem circuit breaker CB 3 and test lamp in Heater No. 1 receptacle shall light.

(2) Place CB 3 in the OFF position. The lamps mentioned in step (1) shall go out.

d. Intercom and convenience circuit breaker (CB 4)

(1) Place Circuit breaker CB 4 in the ON position. The neon lamp above circuit breaker CB 4 and test lamps in Intercom and Convenience 1 & 2 receptacles shall light.

(2) Place CB 4 in the OFF position. The lamps mentioned in step (1) shall go out.

e. Heater No 2 circuit breaker (CB 5) Repeat steps (1) and (2) of Heater No 1 circuit breaker (CB 3) test for CB 5.

f. Equipment circuit breaker (CB 6)

(1) Place circuit breaker CB 6 in the ON position. The neon lamp above CB 6 shall light, and test lamps in the equipment receptacle shall light.

(2) Place circuit breaker CB 6 in the OFF position. The lamps mentioned above in Step (1) shall go out.

g. Main circuit breaker (CB 7)

(1) Place all circuit breakers and switches in the ON position, all lamps shall light and both blowers shall operate.



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(2) Place main circuit breaker in the OFF position. All lamps will go out and the blowers shall cease to operate.

(3) Place all circuit breakers and switches in the OFF position.

4.6.3 Signal (audio) system wiring and components installation tests.-

4.6.3.1 Continuity tests.- The point to point continuity resistance of all signal circuits shall be as described in the following paragraphs. A low resistance continuity test or equivalent shall be used to test continuity resistance.

a. Using the "RADIO-TO-CABLE PATCH THRU" patch cord, connect between the Radio "PATCH THRU SYS-1" and TD-204/754 "PATCH THRU SYS-1" at the patch panel. Repeat for system 2.

b. Continuity resistance of 1 ohm or less shall exist between the following pins of cable assemblies marked "GRC-103 PATCH THRU SYS-1" and "TD-754 PATCH THRU SYS-1": D to F, E to E, K to G, and J to H. Repeat for system 2.

c. Using the "RADIO-TO-CABLE O.W." patch cord connect between the Radio "ORDER WIRE SYS-1" and TD-202/U "ORDER WIRE SYS-1" at the patch panel. Repeat for system 2.

d. Continuity resistance of 1 ohm or less shall exist between the following pins of cable assemblies marked "AN/GRC-103 C to C, and D to D. Repeat for system 2.

e. Using the "12 CH RADIO (GRC-103) REPEATER PATCH THRU" patch cord connect between the Radio "PATCH THRU SYS-1" and Radio "PATCH THRU SYS-2" at the patch panel.

f. Continuity resistance of 1 ohm or less shall exist between the following pins of cable assemblies marked "GRC-103 PATCH THRU SYS-1" and "GRC-103 PATCH THRU SYS-2": D to E, E to G, J to J, and G to E.

g. Using the "CABLE REPEATER PATCH THRU" patch cord connect between the TD-754 "PATCH THRU SYS-1" and TD-754 "PATCH THRU SYS-2" at the patch panel.

h. Continuity resistance of 1 ohm or less shall exist between the following pins of cable assemblies marked "TD-754 PATCH THRU SYS-2": A to E, B to F, E to A, and F to B. Continuity shall also exist between pins C and D of the same cable assembly listed above.

i. Continuity resistance of 1 ohm or less shall exist between the



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U-187 data mux signal connector at the signal entrance box and the audio connectors connecting to the TD-1069/G as shown in Table V.

j. Place all CABLE/BINDING POST channel switches in the BINDING POST position.

k. Connect tester between conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4" and contact 2B of U-187 signal connector, system 1, located at signal entrance box. No circuit continuity shall exist.

l. Place channel switch S1 of system 1 to CABLE position. Circuit continuity shall now exist and shall be 5 ohms or less. Replace channel switch S1 to BINDING POST position.

m. Repeat steps k and l above for each of the remaining conductors of the same audio connector to the corresponding contacts of the same signal connector as shown in Table V; each time placing to CABLE position the channel switch correlating with the circuit being tested and replacing to BINDING POST position following test. (Refer to Signal Schematic for circuit path from audio to signal connectors).

n. Repeat steps k thru m above for audio connectors marked "SYSTEM 1 CHANNELS 5-8" and "SYSTEM 1 CHANNELS 9-12" to the same signal connector as mentioned in step b. above.

o. Repeat steps k thru n above for all remaining audio connectors of system corresponding to system 2 signal connector.

p. Place all CABLE/BINDING POST channel switches in the CABLE position.

q. Connect tester between conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4" and binding post 2B of system 1 located at signal entrance box. No circuit continuity shall exist.

TABLE V

| U-187 Signal Connector CH 1-12 | to | Audio Connector CH 1-4 |
|---|---|---|
| 1A | | D |
| 1B | | C |
| 2A | | B |
| 2B | | A |
| 3A | | M |
| 3B | | L |
| 4A | | K |



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TABLE V (continued)

| Signal Connector CH 1-12 | to | Audio Connector CH 1-4 |
|---|---|---|
| 4B | | J |
| 5A | | H |
| 5B | | G |
| 6A | | F |
| 6B | | E |
| 7A | | S |
| 7B | | R |
| 8A | | P |
| 8B | | N |

| Signal Connector CH 1-12 | to | Audio Connector CH 5-8 |
|---|---|---|
| 9A | | D |
| 9B | | C |
| 10A | | B |
| 10B | | A |
| 11A | | M |
| 11B | | L |
| 12A | | K |
| 12B | | J |
| 13A | | H |
| 13B | | G |
| 14A | | F |
| 14B | | E |
| 15A | | S |
| 15B | | R |
| 16A | | P |
| 16B | | N |

| Signal Connector CH 1-12 | to | Audio Connector CH 9-12 |
|---|---|---|
| 17A | | D |
| 17B | | C |
| 18A | | B |
| 18B | | A |
| 19A | | M |
| 19B | | L |
| 20A | | K |
| 20B | | J |
| 21A | | H |
| 21B | | G |



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TABLE V (continued)

| Signal Connector CH 1-12 | to | Audio Connector CH 9-12 |
|---|---|---|
| 22A | | F |
| 22B | | E |
| 23A | | S |
| 23B | | R |
| 24A | | P |
| 24B | | N |

r. Place channel switch S1 of system 1 to BINDING POST position. Circuit continuity shall now exist and shall be 5 ohms or less.

s. Repeat steps q and r above for each of the remaining conductors of the same audio connectors to be the corresponding binding posts of system 1; each time placing to BINDING POST position the channel switch correlating with the circuit being tested and replacing to CABLE position following test. (Refer to signal schematic for circuit path from audio connector to binding posts.)

t. Repeat steps q thru s above for all remaining audio connectors of system 1 to the corresponding binding posts of system 1.

u. Repeat steps q thru t above for all remaining audio connectors of system 2 corresponding to binding posts of systems 2.

v. Connect tester across the phone binding posts at power entrance box. No continuity shall exist. Repeat for intercom binding posts.

w. Short unterminated phone wires. Connect tester across the phone binding posts at power entrance box. Circuit continuity shall exist and shall be 5 ohms or less. Remove short or unterminated phone wires.

x. Repeat step w above for the intercom wires and binding posts.

y. Connect tester between following points. Circuit continuity shall exist and shall be 5 ohms or less.

(1) From phone A binding post located at power entrance box to:

(a) Binding posts 25A of systems 1 and 2.

(b) Contacts 25A of systems 1 and 2 signal connector.

(2) From phone B binding post located at power entrance box to:



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or equivalent shall be used to test continuity resistance.

(a) Binding posts 25B of systems 1 and 2.

(b) Contacts 25B of systems 1 and 2 signal connectors.

(3) From intercom A binding post located at power entrance box to:

(a) Binding Posts 26A of systems 1 and 2.

(b) Contacts 26A of systems 1 and 2 signal connectors.

(4) From intercom B binding post located at power entrance box to:

(a) Binding Posts 26B of systems 1 and 2.

(b) Contacts 26B of systems 1 and 2 signal connectors.

4.6.3.2 Dielectric breakdown tests.- The dielectric strength of all signal lines shall withstand a potential of 500 volts DC applied continuously for 5 seconds without breakdown. A non-destructive DC dielectric test instrument shall be used in performing the test.

NOTE: Take preventative measures that will preclude personnel from coming into contact with conductors under test during ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test to eliminate danger of shock.

a. Using appropriate test jigs or test fixtures, connect the following to shelter facility ground terminal:

(1) All contacts, 1A thru 26B, of signal connectors, systems 1 and 2 and data mux connector.

(2) All binding posts in the signal and power entrance boxes.

(3) All shields of the six audio connectors of systems 1 and 2 and the three data mux connectors.

(4) All pins of each connector at the PATCH THRU and ORDER WIRE PATCH PANEL.

b. Place all CABLE/BINDING POST channel switches in the CABLE position.

c. Connect tester between shelter facility ground terminal and conductor A of audio connector marked "SYSTEM 1 CHANNELS 1-4". Remove ground con-



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nection form contact 2B of signal connector, system 1.

d. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. Re-establish ground connection(s) to the contact(s) or binding post(s) under test.

e. Repeat steps c and d above for each of the remaining conductors of the same audio connector; each time ungrounding the contact of signal connector correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test. (Refer to the signal schematic.)

f. Repeat steps c thru e above for each of the remaining audio connectors of system 1 and 2 corresponding to signal connectors 1 and 2 respectively and for the data mux connectors.

g. Place all CABLE/BINDING POST channel switches in the BINDING POST position.

h. Repeat steps c thru f substituting binding posts of systems 1 and 2 for contacts of audio connectors 1 and 2 respectively and data mux binding posts for the data mux connectors.

i. Connect tester between shelter facility ground terminal and Phone A binding post at power entrance box. Remove ground connections from contacts 25A of signal connectors 1 and 2; binding posts 25A of systems 1 and 2; and phone A binding post at power entrance box.

j. Repeat step d above with a potential of 250 volts DC output.

k. Repeat steps i and j above for the other Phone and Intercom, binding posts at power entrance box, and shelter facility ground; each time ungrounding all contacts and binding posts correlating with the circuit being tested and reconnecting to shelter facility ground terminal following completion of test.

l. Place tester between Pin C of cable assembly marked RAD HANDSET O.W. SYS-1 and shelter facility ground terminal. Remove ground connection from pin E of AN/GRC-103 PATCH THRU SYS-1 connector at patch panel.

m. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. Re-establish ground connection to the pin under test.

n. Repeat steps l and m above for the following pins of cable assembly RAD HANDSET O.W. SYS-1: E, A, and B; each time ungrounding the pin



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of connector AN/GRC-103 PATCH THRU SYS-1 correlating with the circuit being tested as follows: F, G, and H respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

o. Repeat steps l thru n above substituting system 2 for system 1.

p. Place tester between pin A of cable assembly marked TD-754 PATCH Thru SYS-1 and shelter facility ground terminal. Remove ground connection from pin A of TD-754 PATCH THRU connector at patch panel.

q. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. Re-establish ground connection from pin under test.

r. Repeat steps p and q above for the following pins of cable assembly TD-754 PATCH THRU SYS-1: B, C, D, E, F, G, and H; each time ungrounding the pin of connector TD-754 PATCH THRU SYS-1 correlating with the circuit being tested as follows: B, C, D, E, F, G, and H respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

s. Repeat steps p thru r above substituting system 2 for system 1.

t. Place tester between pin A of cable assembly marked RAD ORDER WIRE SYS-1 and shelter facility ground terminal. Remove ground connection from Pin A of AN/GRC-103 ORDER WIRE connector at patch panel.

u. Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than 5 seconds. Re-establish ground connection to the pin under test.

v. Repeat steps t and u above for the following pins of cable assembly RAD ORDER WIRE SYS-1: B, C, and D: each time ungrounding the pin of connector AN/GRC-103 ORDER WIRE SYS-1 correlating with the circuit being tested as follows: B, C, and D respectively. After each test reconnect the pin being tested to shelter facility ground terminal.

w. Repeat steps t thru v above substituting system 2 for system 1.

x. Repeat steps t thru w above substituting TD-202 ORDER WIRE SYS-1 for RAD ORDER WIRE SYS-1 and TD-202 ORDER WIRE SYS-1 connector for AN/GRC-103 ORDER WIRES SYS-1 connector.

4.6.3.3 Insulation Resistance Tests.- Insulation resistance of all signal lines shall be greater than 100 megohms when tested in accordance with the following paragraphs. A megohmeter tester or equivalent shall be used in performing the tests.



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a. Repeat steps a thru x of paragraph 4.6.3.2 above, except steps d, j, m, q and u which shall be replaced by step b below.

b. Adjust tester to a value of not less than 100 nor more than 500 volts DC output. Maintain this potential for one minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The insulation resistance requirement shall be met. Re-establish ground connection to the conductor under test.

4.6.4 Video (rf) system wiring and components insulation tests.-

4.6.4.1 Continuity tests.- The point to point continuity of all rf cable assemblies shall not exceed 1 ohm. Test procedures shall be as described in the following paragraphs. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test.

a. Place data switch of switchbox to System 1 position.

b. Place system 1 switch of the switchbox to AUDIO position.

c. Place the continuity tester between pin A of both system 1 10-pair connectors at the switchbox. The resistance reading shall be less than 1 ohm.

d. Repeat step b for each remaining pin of the same connectors except pins U, V and T.

e. Replace switch in the DATA position.

f. Repeat steps c and d except that no continuity shall exist between the following pins: E, F, G, and H.

g. Place tester between inner conductor of switch box DATA OUT SYS-1 connector and pin H of switchbox MUX CH 5-8 SYS-1 connector. Resistance reading shall be less than 1 ohm.

h. Repeat step g for the shield of DATA OUT SYS-1 connector and pin G of switchbox MUX CH-5-8 SYS-1 connector.

i. Repeat step g for the inner conductor of switchbox DATA IN SYS-1 connector and pin F of switchbox MUX CH 5-8 SYS-1 connector.

j. Repeat step g for the shield of DATA IN SYS-1 connector and pin E of switchbox MUX CH 5-8 SYS-1 connector.

k. Repeat steps g thru h with the switch in the AUDIO position. No continuity shall exist.



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l. Place data switch of switchbox to System 2 position.

m. Repeat steps b thru l for system 2.

n. Connect a low resistance reading ohmmeter between the following cable assemblies and patch panel. Method of testing shall be from inner conductor to inner conductor and from outer conductor (shield) to outer conductor (shield). Continuity resistance shall not exceed 1 ohm.

| FROM PATCH PANEL MARKED | TO CABLE ASSEMBLY MARKED |
|---|---|
| TD-660 PCM OUT SYS-1 | TD-660 PCM OUT SYS-1 |
| TD-660 PCM IN SYS-1 | MUX PCM IN SYS-1 |
| TD-660 TIM IN SYS-1 | MUX TIM IN SYS-1 |
| TD-660 TIM OUT SYS-1 | MUX TIM OUT SYS-1 |
| TD-660 ALT PCM OUT SYS-1 | ALT PCM OUT SYS-1 |
| TD-660 SYNC OUT XMTR SYS-1 | SYNC OUT XMTR SYS-1 |
| TD-660 SYNC IN SYS-1 | SYNC IN SYS-1 |
| TD-660 SYNC OUT RCVR SYS-1 | SYNC OUT RCVR SYS-1 |
| TD-202 PCM OUT SYS-1 | TD-202 PCM OUT-1 SYS-1 |
| TD-202 PCM IN SYS-1 | TD-202 PCM IN-1 SYS-1 |
| TD-202 TIM IN SYS-1 | TD-202 TIM IN SYS-1 |
| TD-202 TIM OUT SYS-1 | TD-202 TIM OUT-1 SYS-1 |
| TD-202 PCM IN-2 SYS-1 | TD-202 PCM IN-2 SYS-1 |
| TD-202 PCM OUT-2 SYS-1 | TD-202 PCM OUT-2 SYS-1 |
| TD-202 TIM OUT-2 SYS-1 | TD-202 TIM OUT-2 SYS-1 |
| TD-754 PCM OUT SYS-1 | COMB PCM OUT-1 SYS-1 |
| TD-754 PCM IN SYS-1 | COMB PCM IN-1 SYS-1 |
| TD-754 TIM IN SYS-1 | COMB TIM IN SYS-1 |
| TD-754 TIM OUT SYS-1 | COMB TIM OUT-1 SYS-1 |
| TD-754 PCM IN-2 SYS-1 | COMB PCM IN-2 SYS-1 |
| TD-754 PCM OUT-2 SYS-1 | COMB PCM OUT-2 SYS-1 |
| TD-754 TIM OUT-2 SYS-1 | COMB TIM OUT-2 SYS-1 |
| TD-754 FROM CABLE SYS-1 | TD-754 FROM CABLE SYS-1 |
| TD-754 TO CABLE SYS-1 | TD-754 TO CABLE SYS-1 |
| VID ENT FROM CABLE SYS-1 | VID ENT BOX MALE(M) SYS-1 |
| VID ENT TO CABLE SYS-1 | VID ENT BOX FEMALE(F) SYS-1 |
| AN/GRC-103 RADIO IN SYS-1 | RAD PCM IN SYS-1 |
| AN/GRC-103 RADIO OUT SYS-1 | RAD PCM OUT SYS-1 |
| TD-202 RADIO IN SYS-1 | TD-202 FM RAD RCVR SYS-1 |
| TD-202 RADIO OUT SYS-1 | TD-202 to RAD XMTR SYS-1 |

o. Repeat continuity check of step n above substituting system 2 for system 1.

p. Measure continuity resistance between the inner conductor of antenna connector, system 1, and inner conductor of cable assembly marked to



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XMTR of system 1. The resistance reading shall not exceed 1 ohm.

q. Repeat step p above for the shields of the same circuit.

r. Repeat steps p and d above substituting system 2 for system 1.

4.6.4.2 Dielectric and insulation resistance tests.- Dielectric and insulation resistance tests shall be performed on each video and rf circuit in accordance with the following paragraphs. A non-destructive dielectric and insulation resistance tester shall be used in performing the test.

a. At Cable Patch Panel make the following cable connections using Cable Patch Cords:

| | |
|---|---|
| VIDEO ENT BOX SYS-1 | TD-754 SYS-1 |
| FROM CABLE | FROM CABLE |
| TO CABLE | TO CABLE |
| VIDEO ENT BOX SYS-2 | TD-754 SYS-2 |
| FROM CABLE | FROM CABLE |
| TO CABLE | TO CABLE |

b. Remove all lightning protector elements at video entrance box.

c. Apply test potential of 1500 volts DC for not less than 30 seconds between the following pin connections and terminal ground located at video entrance box. There should be no breakdown and insulation resistance shall not be less than 100 megohms.

(1) System 1 male

(2) System 1 female

(3) System 2 male

(4) System 2 female

d. Replace lightning protector elements removed by step b above.

e. Repeat step c above applying test potential of 1100 volts DC.

f. Connect tester between inner conductor of antenna connector, system 1, and shelter facility ground terminal.

g. Adjust tester to 1700 RMS output and maintain this potential for a continuous period of not less than 30 seconds. There shall be no breakdown and the insulation resistance shall be less than 100 megohms.



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h. Repeat steps f and g above substituting system 2 for system 1.

i. Apply test potential of 1500 volts DC for not less than 5 seconds between inner and outer (shield) conductor of each RF connector at the Video and Radio Patch Panel. There shall be no breakdown and the insulation resistance shall not be less than 100 megohms.

j. Place both system switches of Switch Box to AUDIO position.

k. Place Data switch in System 1 position.

l. Connect tester between each pin of the switchbox SWPD CH 5-8 Sys 1 connector and all other pins and shield of the same connector connected together and to facility ground.

m. Apply test potential of 1500 volts DC for not less than 30 seconds. There shall be no breakdown and insulation resistance shall not be less than 100 megohms.

n. Repeat steps k thru m above for system 2.

o. Connect tester between inner conductor and shield of each rf connector at the switchbox.

p. Repeat step m above for each rf connector.

4.7 Systems operation test.- Install all operational equipments in their respective rack locations and connect all power, signal (audio) and rf cable assemblies to their designated receptacles and connectors.

4.7.1 12 Channel Radio/Cable terminal test.- (See Figure 1 for block diagram test setup.)

a. At the mux patch panel make the following cable connections using "mux patch cords":

| | | |
|---|---|---|
| TD-660 SYS-1 | to | AN/GRC-103 SYS-1 |
| 12/24 CH PCM IN | | 12 CH PCM OUT |
| 12 CH PCM OUT | | 12/24 CH PCM IN |
| TIM IN | | TIM OUT |
| TD-660 SYS-2 | to | TD-754 SYS-2 |
| 12 CH PCM OUT | | 12/24 CH PCM IN |
| 12/24 CH PCM IN | | 12 CH PCM OUT |
| TIM OUT | | TIM IN |
| TIM IN | | TIM OUT |



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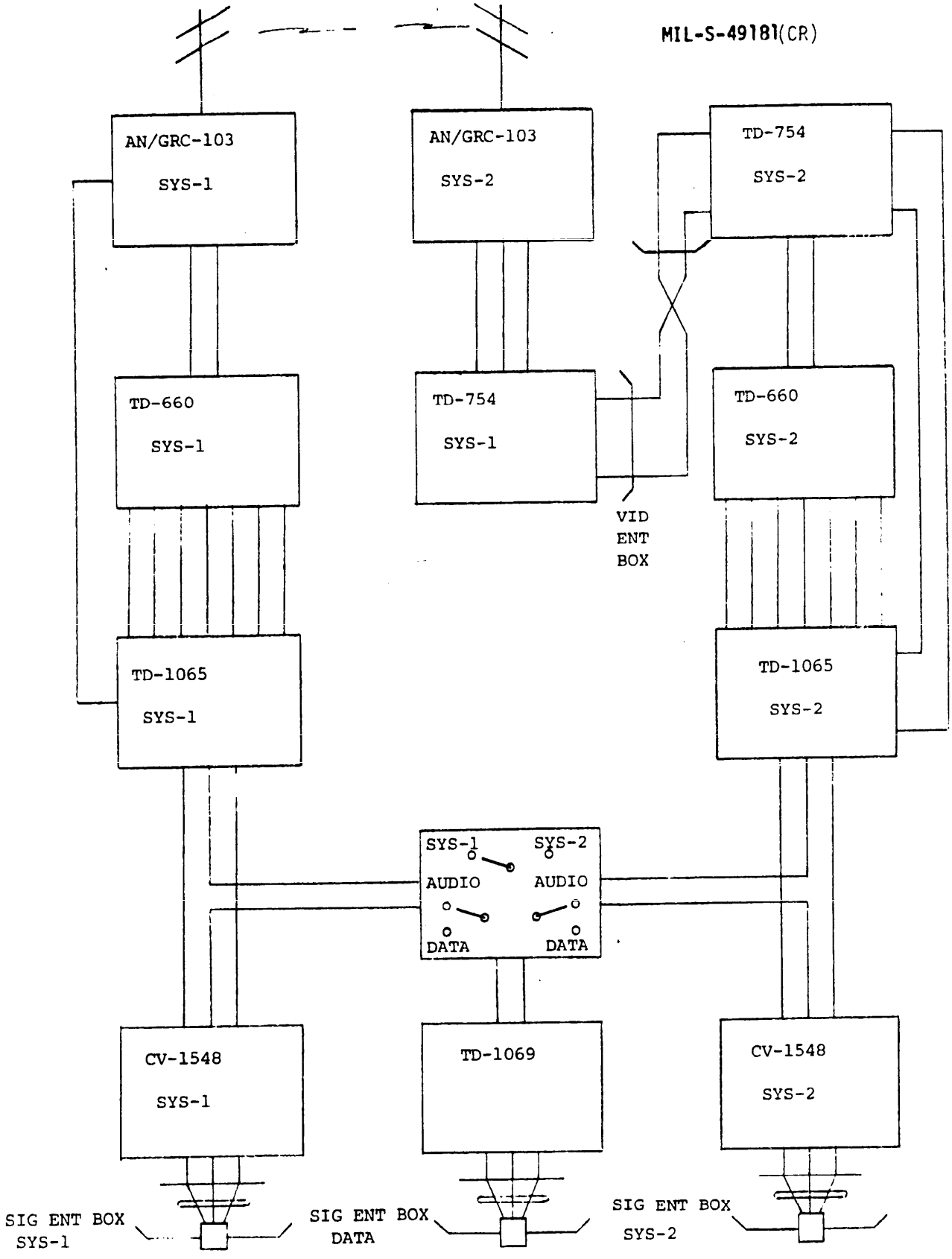



FIGURE 1 TWELVE CHANNEL TEST SET-UP



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| TM-754 SYS-1 | to | AN/GRC-103 SYS-2 |
|---|---|---|
| 12/24 CH PCM IN | | 12 CH PCM OUT |
| 12 CH PCM OUT | | 12/24 CH PCM IN |
| TIM IN | | TIM OUT |

b. At the order Wire Patch Panel make the following cable connection using the "RADIO-TO-CABLE PATCH THRU" patch cord: AN/GRC-103 PATCH THRU system 2 to TD-754 PATCH THRU system 1.

c. At the cable patch panel make the following cable connections using "cable patch cords":

| TD-754 SYS-1 | to | VID ENT BOX SYS-1 |
|---|---|---|
| TO CABLE | | TO CABLE |
| FROM CABLE | | FROM CABLE |
| **TD-745 SYS-2** | | **VID ENT BOX SYS-2** |
| TO CABLE | | TO CABLE |
| FROM CABLE | | FROM CABLE |

d. Interconnect system 1 and system 2 video connectors at video entrance box using cable assembly CX-11230/G.

e. Connect Cable Assemblies XC-1859/U (40 ft) or CG-3443/U (80 ft) to each antenna connector at antenna entrance box. Connect other ends to Cable Assemblies GC-718( )/U (6 ft) or CG-3444/U (3 ft 6 in) using Adapter UG-1275 ( )/U. Connect other end of GC-718( )/U or CG-3444/U to Dummy Loads DA-437/GRC. Separate the dummy loads to a distance of about 2 feet.

f. Connect Cable Assemblies CX-4566/U (25 ft) to signal connecors "SYS-1 CH 1-2" and "SYS-2 CH 1-12". Connect other ends of the cable assemblies to the two Junction Boxes J-1077A/U.

g. Operate both systems switches of the switch box to the "AUDIO" position; and all channel switches on signal entrance panel to "BINDING POST".

h. Perform the following operational checks and adjustments of Telephone Signal Converters CV-1548/G.

(1) Check to see that the correct plug-in panels are seated firmly in their slots.

(2) In the power supply panel 18A1 check to see that the correct protective fuse (1 1/2 amp) is installed in the fuseholder.



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(3) Operate the 2W/4W channel switches to 2W.

(4) Operate the power switch to ON. The power indicator lamp shall light and the 20 neon lamp on panel 18A2 shall glow.

(5) Operate Signalling mode switch on each panel 18A3 to OFF.

(6) Operate the meter selector switch through each of the following positions and observe that the TEST - ALIGN meter indicates as listed below:

| Switch Position | Meter Indication |
|---|---|
| - | Yellow area |
| + | Yellow area |
| 20 Drive | Yellow area |
| 20 | Yellow area |
| 1600 | Green area (if not, adjust ADJ 1600 control on panel 18A2.) |

(7) If the TEST - ALIGN meter does not provide proper indication as listed in step (5) above, perform correction indicated:

| Normal Indication | Corrective Action |
|---|---|
| - (yellow) | Replace 1/2 ASB fuses on panel 18A1 |
| + (yellow) | Replace 1 1/2A fuse panel 18A1 |
| 20 Drive (yellow) | Replace panel 18A2 |
| 20 (yellow) | Replace panel 18A2 |
| 1600 (green) | Replace panel 18A2 |

i. Perform the following preliminary procedures on both Multiplexers TD-660A/G:

(1) Check to see that the correct plug-in panels are seated firmly in their proper slots.

(2) Operate switches I, II, III and IV to the OFF position.

(3) Operate the AUX switch to OUT.

(4) Operate the MASTER/SLAVE switch to MASTER.

(5) Operate the 2W-4W switch to 4W.



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(6) Operate the 6CH/12CH switch the 12CH.

(7) Operate the AC power switch to ON. The power indicator lamp shall light. Ignore the frame alarm lamps and the audible alarm. To silence the alarm depress the BUZZER OFF switch.

(8) Operate the selector switch I to each of the positions listed below and observe the TEST ALIGN meter.

| Switch Position | Meter Indication |
|---|---|
| +7 | Yellow |
| +12 | Hairline |
| +4 | Green |
| -12 | Green |
| -6 | Green |
| -4 | Green |

(9) If the TEST ALIGN meter indications do not correspond to the above, perform the +12 volt adjustment as follows and if the meter indications are still incorrect, replace applicable fuse or panel.

(a) Operate selector switch I to +7 and observe that the TEST ALIGN meter indicates yellow.

(b) Operate selector switch I to +12.

(c) Loosen the locknut on the +12V control.

(d) Adjust the +12V control for a hairline reading on the TEST ALIGN meter.

(3) Tighten the -12V control locknut.

(10) Operate selector switch I to the SWIII position.

(11) Operate selector switch III to the 060 position. The TEST ALIGN meter should indicate the green area, signifying the tone oscillator is functioning properly. If the TEST ALIGN meter does not indicate in the green area replace panel 11A8.

j. Perform the following procedures on Multiplexer TD-1069/G:

(1) Operate the 188/TTL switch to 188 position.

(2) Operate all 24 PORT MODE SELECT switches to 0.6 position.



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(3) Operate the MODE switch to 12.

(4) Operate the NORM OPR - ZERO Set - READ switch to NORM OPR.

(5) Operate both MILE switches to 1/4.

(6) Operate the TONE-OFF switch to OFF.

(7) Connect headsets to Headset connectors.

(8) Operate the AC POWER switch to ON. Check to see that the AC POWER and ALARMS NO CABLE CUR indicators light and the buzzer sounds. To silence the alarm depress BUZZER OFF switch.

m. Perform the following preliminary procedures on each Multiplexer TD 202/U:

(1) Operate OPR/TEST switch to OPR, and TRAFFIC SEL switch to 12.

(2) Operate the AC POWER switch to ON. The AC power indicator shall light and the buzzer on the TRAFFIC alarm shall sound. To silence the alarm depress the BUZZER OFF switch.

n. Perform the following preliminary, tuning, and operating procedures on each Radio Set AN/GRC-103(V)4:

(1) Set the transmitter XMTR CHANNEL and XMTR TUNE controls to the assigned transmitter channel number.

(2) Set the receiver XMTR CHANNEL control to the assigned transmitter channel number.

(3) Set the receiver RCVR CHANNEL and RCVR SIG controls to the receiver channel number. (Separate receiver and transmitter by at least 40 channels.)

(4) Set the transmitter AC POWER switch to ON/RESET and verify that the centrigual fan starts. AC POWER lamp lights, LOW POWER lamp lights, SYNC lamp may light momentarily (lamp goes out within 10 seconds), and buzzer may sound (press BUZZER OFF pushbutton to silence buzzer).

(5) Set receiver AC POWER switch to ON and verify that the AC POWER lamp lights, order wire until POWER lamp lights, LOW SIGNAL lamp lights, SYNC lamp may light momentarily (lamp goes out within 10 seconds), buzzer may sound (press BUZZER OFF to silence buzzer), and loud rushing sound is heard in order wire unit handset.



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(6) Set transmitter meter switch to 12 VDC. Meter shall read in the green band.

(7) Set transmitter meter switch to 28 VDC. Meter shall read in the green band.

(8) Set transmitter meter switch to 600 VDC. Meter shall read in the green band.

(9) Set transmitter switch to OSC. Meter shall read between 25% and 90% of full scale.

(10) Set transmitter switch to DOUBLE. Meter shall read between 25% and 90% of full scale.

(11) Set transmitter switch to MULT. Meter shall read between 25% and 90% of full scale.

(12) Set transmitter meter switch to DRIVER. Tune PWR OUT PEAK control for maximum meter reading. Meter shall read between 25% and 90% of full scale.

(13) Set transmitter switch to PWR OUT. Tune PWR OUT PEAK control for maximum meter reading. Meter shall read between 25% and 90% of full scale.

(14) Set transmitter meter switch to REFL PWR. Tune receiver XMTR SUPL control for minimum transmitter reading. Check to see that receiver XMTR CHANNEL indicator is within 10 channels of the assigned transmitter channel number.

(15) Set receiver meter switch to XMTR DUPL. Meter shall read between 25% and 90% of full scale.

(16) Set receiver meter switch to REFL PWR. Meter shall read less than 20% of full scale.

(17) Set transmitter meter switch to 12 CH PCM. Adjust ransmitter input control until transmitter meter indicates in green band.

(18) Set receiver meter switch to -12 VDC. Meter shall read to green band.

(19) Set receiver meter switch to -12 VDC. Meter shall read in green band.

(20) Wait until SYNC lamp goes out. NOTE: The sync light may light momentarily, or may not light at all.



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(21) Set receiver meter switch to OSC. Meter shall read between 25% and 90% of full scale.

(22) Set receiver meter switch to DOUBLER. Meter shall read between 25% and 90% of full scale.

(23) Set receiver meter switch to MULT. Adjust MULT PEAK control for full maximum meter reading. Meter shall read between 25% and 90% of full scale. LOW SIGNAL lamp will go out if distant transmitter is operating properly. (NOTE: Move the dummy loads accordingly to improve the receive signal. If the signal strength is still low remove one dummy load and leave cable assembly unterminated. If signal strength is still low remove the remaining dummy load and position unterminated cable assemblies to assure a good signal strength.) The loud rushing sound in the order wire handset shall disappear when the LOW SIGNAL lamp goes out. The buzzer may sound. Push BUZZER OFF button to silence buzzer.

(24) Set meter switch to RCVR SIG. Meter shall read between 25% and 90% of full scale.

(25) Push RING button on radio order wire unit of system 1. A 1600 Hz signalling tone is heard at the O.W. handset and at both TD-754's. Communicate between the O.W. handset and TD-754 headsets.

(26) Repeat step (25) signalling from system 2 O.W.

o. All equipment shall be operational. To silence the alarms depress the BUZZER OFF switches. If normal indications are not encountered during the following operational procedures, do not continue until proper indications are obtained. If proper indications cannot be obtained by adjusting or replacement of panels, replace the equipment.

p. The following operating procedures apply to both TD-202/U's.

(1) Operate METER SELECT switch to the following positions and check for proper indications on the TEST ALIGN meter.

| POSITION | INDICATION |
|---|---|
| TIM IN | GREEN |
| PCM IN | GREEN |
| TO RADIO XMTR* | HAIRLINE |

*Adjust OL on panel 5A2 for hairline indication on TEST ALIGN METER, with distant TD-202/U set to FROM RADIO RCVR. Local radio must adjust PCM level



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for proper indications. Distant TD-202/U must adjust RL and C. on panel 5A3 for green area indication and peak indication respectively.

| POSITION | INDICATION |
|---|---|
| FROM RADIO RCVR** | HAIRLINE |

** Adjust RL on panel 5A3 for green area and CL on panel 5A3 for peak indication, with distant TD-202/U set to TO RADIO XMTR. Distant TD-202/U must adjust OL on panel 5A2 for hairline indication. Distant radio must adjust PCM level for proper indications.

(2) Operate SERV SEL to O. At distance radio transmitter set multimeter selector switch to 1 KC MOD (test tone). Adjust ORDER WIRE control knob until TEST ALIGN indicates in yellow area. Turn distance radio test tone off.

(3) Repeat step (2) with distant TD-202/U and local radio transmitter test tone on. After adjustment turn test tone off.

(4) Set SERV SEL switch to the positions and observe TEST ALIGN meter for proper indications. Make necessary voltage adjustments.

| SWITCH POSITION | METER INDICATION |
|---|---|
| +10 | HAIRLINE |
| +4.5 | HAIRLINE |
| -4.5 | HAIRLINE |

(5) Set SERV SEL switch to the following positions. If TEST ALIGN meter does not indicate in green area, replace appropriate panel.

A Panel 5A3 (Input from radio present and RL adjusted)
B Panel 5A4
C Panel 5A5 (CL adjusted; may not be in green area)
D Panel 5A5
C Panel 6A7 (TIMING OUT-1)
D Panel 6A7 (TIMING OUT-2) (24 channel operation only)
E Panel 6A7 (PCM OUT-1)
F Panel 6A7 (PCM OUT-2)
G Panel 6A7 (Osc Bins)
H Panel 6A5 (Transmit PCM)
J Panel 6A5 (Transmit Timing)
K Panel 6A6 (Mixer In-1)
L Panel 6A6 (Mixer In-2)
M Panel 6A6 (Mixer Output)
N Panel 6A6 (Reset Timing)



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O Panel 6A3 (Cable Transmit Order Wire)
(Tone switch on panel 6A2 must be on and TALK OFF SIG Switch must be in OFF position.)
P Panel 6A3 (Radio Transmit Order Wire)
(Tone switch on panel 6A2 must be on; PATCH THRU connected; and TALK OFF SIG switch in OFF position.)
O Panel 6A2 (Phone Amplifier)
S Panel 6A7 (Activity)

(6) Operate SERV SEL switch to +10 and leave in this position during operation.

q. The following operating procedures apply to both TD-745/G's.

(1) Operate METER SELECT switch to the following positions and check for proper indications on TEST ALIGN meter:

| Position | Indication |
|---|---|
| TIM IN | Green |
| PCM IN-1 | Green |

(2) Operate METER SELECT switch to SERV FAC.

(3) Perform the following order wire communication checks and alignments:

(a) Connect headset or handset to the HEADSET connector of each combiner.

(b) Momentarily operate the TALK OFF SIG switch on one combiner to SIG. The CALL indicator shall light and the buzzer shall sound on the other combiner. Place TALK OFF SIG switch to OFF.

(c) Repeat step (b) reversing the process.

(d) Place TALK OFF SIG switches of both combiners to TALK. Communicate between the headsets or handsets connected to the combiners.

(e) Operate the SERV SEL switch to "O".

(f) Operate TONE OFF switch, located on panel 12A2, of both combiners to TONE.

(g) Adjust CRL potentionmeter, located on panel 12A3, of both combiners to a mid-scale green TEST ALIGN meter indication.



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(h) Operate the TONE OFF switch to OFF.

(4) Operate the SERV SEL switch to the following positions and check for hairline indication on TEST ALIGN meter. Do not continue until proper indications are obtained. If adjustments are necessary set appropriate control for hairline indication on TEST ALIGN meter. If proper indications cannot be obtained, replace appropriate fuse.

| | | |
|---|---|---|
| +28V | Subassembly 12A1 | |
| +12V | Subassembly 12A1 | (+12V ADJ) |
| +4V | Subassembly 12A1 | (+4V ADJ) |
| -6V | Subassembly 12A1 | (-6V ADJ) |

(5) Operate SERV SEL switch to the following positions. If TEST ALIGN meter does not indicate in green area, replace appropriate panel. Do not continue until proper indications are obtained.

| | |
|---|---|
| A | Panel 12A4 |
| B | Panel 12A4 |
| C | Panel 12A4 |
| D | Panel 12A4 |
| E | Panel 12A4 |
| F | Panel 12A4 |
| G | Panel 12A4 |
| H | Panel 12A4 |
| J | Panel 12A4 |
| K | Panel 12A4 |
| L | Panel 12A4 |
| M* | Panel 12A3 |
| O | Panel 12A2 |

*TEST TONE switch to ON.

r. The following operating procedures apply to Both TD-660/U's.

(1) Perform the following metering checks:

| Switch Position | Meter Indication |
|---|---|
| PCM IN | Green |
| TIM IN | Green |
| NOISE GEN | Yellow |

(2) Operate selector switch I to SWII. Operate selector switch II to the following positions. If TEST ALIGN meter does not indicate in green area, replace appropriate panel. Do not continue until proper indications are obtained.



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A Panel 11A2
B Panel 11A3
C Panel 11A2
D Panel 11A3
E Panel A5 (At least one output channel modulated)
F Panel 11A4
G Panel 11A2
H Panel 11A7
J Panel 11A6 (Unit should be out of frame)
K Panel 11A7
L Panel 11A6 (At least one input channel modulated)

(3) Perform the following automatic gain level check:

(a) Operate selector switch I to the SWII position.

(b) Operate selector switch III to the MEAS position.

(c) Operate selector switch IV of both units simultaneously to each channel position (1 through 12) and observe the TEST ALIGN meter for hairline indications.

(d) At each misadjusted channel, adjust the AG control until a hairline indication is observed on the TEST ALIGN meter. NOTE: The channel AG controls are located on the face panel of the three 11A5 panels. Channel controls are identified by numbers stenciled on the perforated covers. The top line of numbers applies to 12-channel operation; the bottom line applies to 6-channel operation. The top AG control on the 11A5 panels represents the highest-numbered channel.

(e) If the AG control does not adjust the automatic gain level to hairline, replace appropriate 11A5 panel.

(4) Perform the following two-way speech test:

(a) Operate all selector switches to OFF.

(b) Connect Handset H-156/U to the TALK MONITOR receptacle of each TD-660/G.

(c) Operate selector switch III to TALK.

(d) Starting at the channel 1 position of selector switch IV, perform a two-way speech test for each channel in sequence, observing that each channel gives satisfactory speech quality.

s. Perform the following communications operational check:



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(1) Connect a Telephone Set TA-312/PT to pair 2WT/R, CH-1 of System 1 and System 2 binding posts. Establish two way voice communication.

(2) Repeat step (1) for channels 2-12.

(3) Place all CV-1548/G 2W/4W channel switches in the 4W position.

(4) Connect a 4 wire Telephone Set TA-838/TT to CH-1 binding posts of System 1 and System 2 establish two way voice communication.

(5) Repeat step (4) above for Ch-2 thru 12.

t. Perform the following verification tests using the provided data generator:

(1) Place all channel switches at the Signal entrance panel in the "cable" position.

(2) On junction box connected to system 1 CH 1-12 short all even "a" pairs (2A, 4A, ... thru 24A) and all even "b" pairs (2B, 4B, ... thru 24B). This represents all data in (transmit) channels (1-12) into system 1, channels, 1-12.

(3) At the junction box with the shorted even pairs connect the output CDP (Conditioned Diphase Polar) signal from the data operator to channel 1 transmit of the junction box (Pair 2).

(4) Connect the input CDP signal of the data generator to channel 1 receive of the other junction box (pair 1).

(5) Set data generator DATA RATE to 16,000 (16 kb/s). DATA switch to RUN, and multiswitch to ERROR.

(6) Turn data generator ON. Panel meter shall indicate no data errors.

(7) Manually insert a single error by depressing the SINGLE ERROR INSERTER pushbutton switch once. Panel Meter shall show one error indication.

(8) Reset COUNTER to zero errors transmitted.

(9) Repeat steps (5) thru (8) with DATA RATE set to 32,000 (32 kb/s).

(10) Repeat steps (3) thru (9) for CH 2 thru 12 retaining data generator ON and the CDP input signal connected to each remaining odd pair thru pair 23 of the same junction box.



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(11) Repeat the test of steps (3) thru (10) with the following conditions.

(a) Junction box with shorted even pairs connected to system 2 at the signal entrance box.

(b) Remaining junction box connected to system 1 at the signal entrance box.

u. Perform the following verification tests using the provided data generator:

(1) Connect the junction box with the shorted even pairs to the Data Mux connector at the Signal Entrance Box.

(2) Connect an external TD-1069/G to System 2 signal connector at the signal entrance box as shown in Fig-2.

(3) Operate the 188/TTL switch of the external TD-1069/G to the 188 position.

(4) Operate all 24 Port Mode Select switches to the 0.6 position.

(5) Operate the power ON/OFF switch to ON. The power indicator lamp shall light. Ignore the frame alarm lamp and the audible alarm. To silence the alarm depress the Audible Alarm Disable switch.

(6) Place the Data switch of the data mux switch in the "SYSTEM 1" position. Place the System 1 switch in the "Data" position, and place the System 2 switch in the "Audio" position.

(7) At the junction box with the shorted even pairs connect the output of the CDP (Conditioned Diphase Polar) signal from the data generator to Channel 1 Transmit of the junction box (pair 2).

(8) Connect the input CDP signal of the data generator to Channel 1 receive of the other junction box (see Fig 2).

(9) Set data generator DATA RATE to 600, Data Switch to RUN, and multiswitch to ERROR.

(10) Turn data generator ON. Panel meter shall indicate no data errors.

(11) Manually insert a single error by depressing the SINGLE ERROR INSERTOR pushbutton switch once. Panel meter shall show one error indication.



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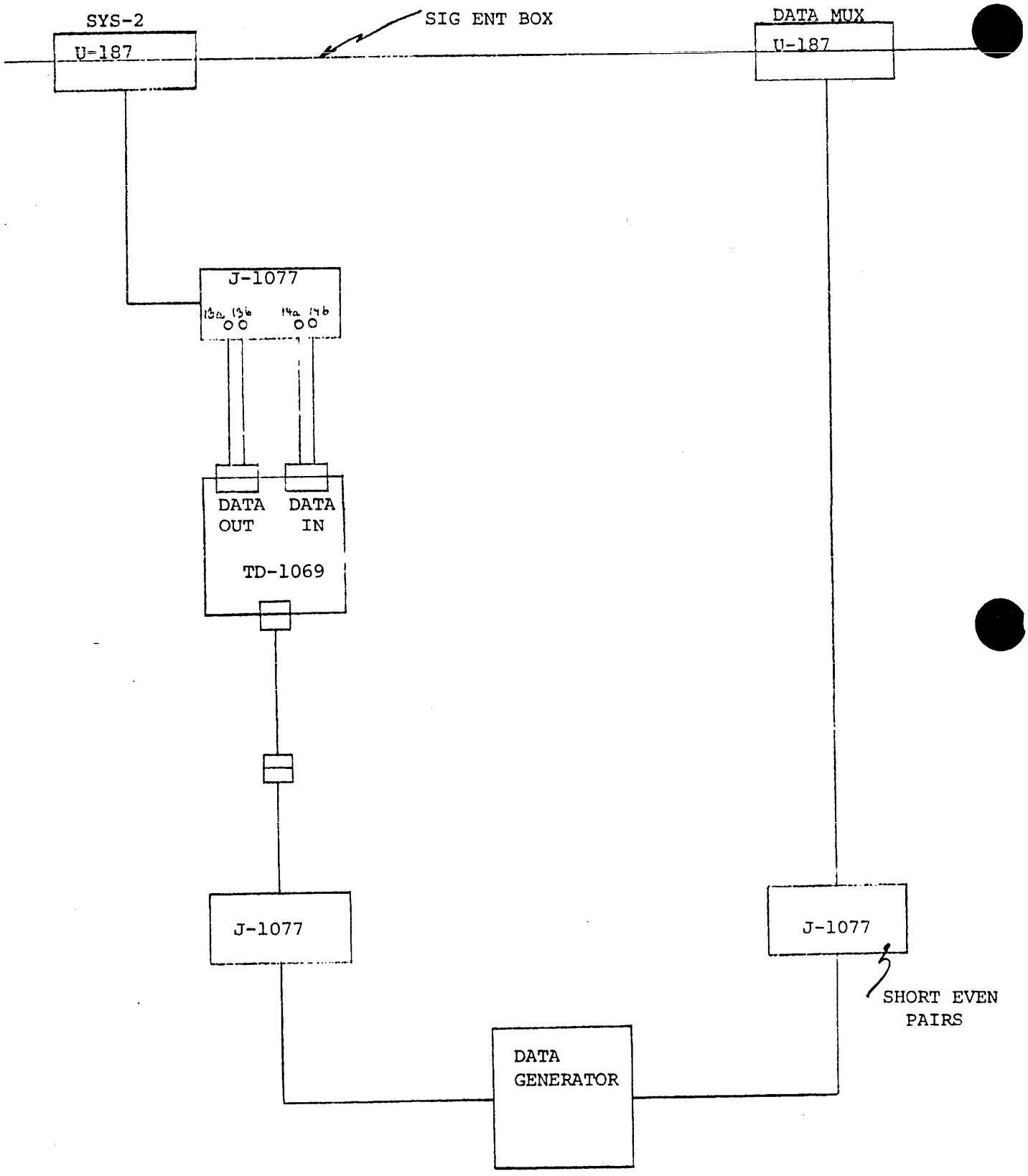



FIGURE 2

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(12) Reset counter to zero errors transmitted.

(13) Repeat steps (7) thru (12) above for CH 2-12 with DCP input signal connected to each remaining odd pair thru pair 223 of the same

(14) Repeat steps (7) thru (13) above with the data generator set at 1200 DATA RATE and all PORT MODE switches of both TD-1069/G set to 1.2.

(15) Connect the junction box with the shorted even pairs to the external TD-1069/G and connect the other junction box to the data mux connector at the Signal Entrance Box.

(16) Place all PORT MODE Select switches of both TD-1069/G to 0.6.

(17) Repeat steps (7) thru (14) above.

(18) Place the data switch of the data mux switch box in the "System 2" position, place the System 1 switch in the "Audio" position, place the System 2 switch in the "Data" position.

(19) Connect the junction box with the shorted even pairs to the data mux connectors at the Signal Entrance box and connect the external TD-1069/G to System 1.

(20) Repeat steps (7) thru (17) above.

4.7.2 24 Channels Radio-To-Cable conversion operational check.- (See Figure 3 for block diagram test setup.)

a. Position TD-660's or TD-754 outside the AN/TRC-151 so as to facilitate connections to the video entrance box.

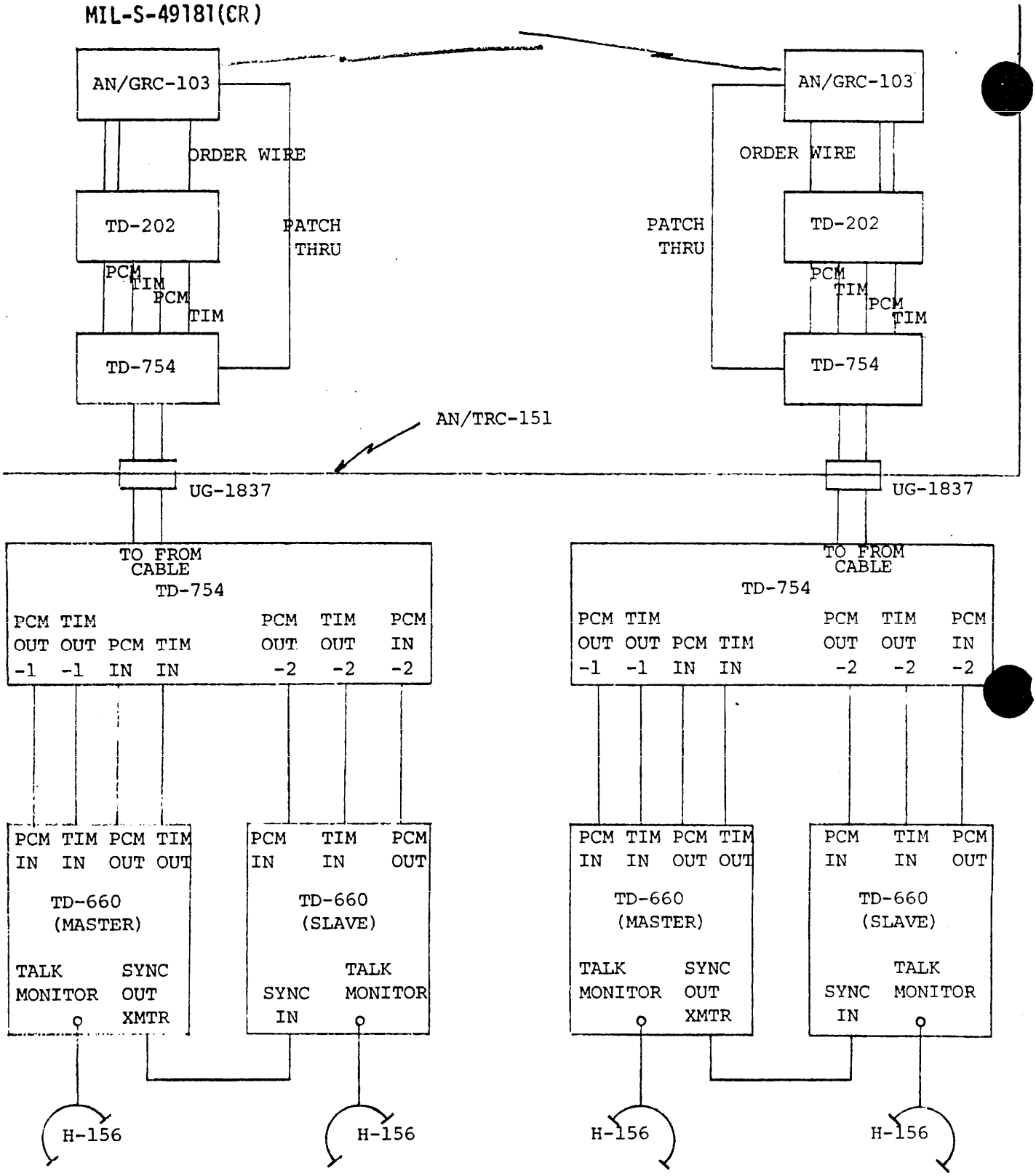
b. Fabricate two cable assemblies to interconnect the external TD-754's with the video entrance box connectors. The cable assemblies shall contain one UG-1837 connector at tone end and a female and male mhv connectors at the other end. The male pin at the UG-1837 shall be connected to the female mhv connector; the female of the UG-1837 shall be connected to the male mhv connector.

c. Using one of the fabricated cable assemblies interconnect an external TD-754 to SYS-1 video entrance connector. Repeat for SYS-2 video connector using the remaining external TD-754.

d. Interconnect external TD-754's and TD-660's as shown in Figure 3.



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IGURE 3 TWENTY-FOUR CHANNEL TEST SET-UP



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e. At the mux patch panel make the following cable connections using "mux patch cords".

| TD-754 SYS-1 | TO | TD-202 SYS-1 |
|---|---|---|
| 12/24 CH PCM IN | | 12 CH PCM OUT |
| 12 CH PCM OUT | | 12/24 CH PCM IN |
| TIM OUT | | TIM IN |
| TIM IN | | TIM OUT |
| PCM IN-2 | | PCM OUT-2 |
| PCM OUT-2 | | PCM IN-2 |

| TD-754 SYS-1 | TO | TD-202 SYS-2 |
|---|---|---|
| 12/24 CH PCM OUT | | 12 CH PCM OUT |
| 12 CH PCM OUT | | 12/24 CH PCM IN |
| TIM OUT | | TIM IN |
| TIM IN | | TIM OUT |
| PCM IN-2 | | PCM OUT-2 |
| PCM OUT-2 | | PCM IN-2 |

f. At the cable patch panel make the following cable connections using "cable patch cords".

| TD-754 SYS-1 | TO | VID ENT BOX SYS-1 |
|---|---|---|
| TO CABLE | | TO CABLE |
| FROM CABLE | | FROM CABLE |

| TD-754 SYS-2 | | VID ENT BOX SYS-2 |
|---|---|---|
| TO CABLE | | TO CABLE |
| FROM CABLE | | FROM CABLE |

g. At the radio patch panel make the following cable connections using "radio patch cords."

| AN/GRC-103 SYS-1 | TO | TD-202 SYS-1 |
|---|---|---|
| 12/24 CH PCM IN | | 24 CH RAD PCM OUT |
| 24 CH RAD PCM OUT | | 24 CH RAD PCM IN |

| AN/GRC-103 SYS-2 | TO | TD-202 SYS-2 |
|---|---|---|
| 12/24 CH PCM IN | | 24 CH RAD PCM OUT |
| 24 CH RAD PCM OUT | | 24 CH RAD PCM IN |



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h. At the order wire patch panel make the following cable connections using the "radio-to-cable patch thru" patch cords: AN/GRC-103 Patch Thru SYS-1 to TD-754 Patch Thru SYS-1; and AN/GRC-103 Patch Thru SYS-2 to TD-754 Patch Thru SYS-2.

i. At the order wire patch panel make the following cable connections using the "radio-to-cable order wire" patch cords: AN/GRC-103 Order Wire SYS-1 to TD-202 Order Wire SYS-1; and AN/GRC-103 Order Wire SYS-2 to TD-202 Order Wire SYS-2.

j. Connect Cable Assemblies G-1859/u (40 ft) or CG-3443/u (80 ft) to ANT-1 and ANT-2 connectors at the antenna entrance box. Connect other ends to Cable Assemblies CG-718( )/G (6 ft) or CG-3444/u to Dummy Loads DA-437/GRC. Separate the dummy loads to a distance of about 5 feet. NOTE: During the radio tuning procedures it might be necessary to move the dummy loads accordingly to receive a suitable signal, or it might be necessary to completely disconnect both dummy loads and adjust the ends of the unterminated cable assemblies until a good signal is received.

k. Perform the following two-way speech test:

(1) Operate all selector switches on the TD-660 to OFF.

(2) Connect Handset H-156/U to the TALK MONITOR receptacle of each TD-660.

(3) Operate selector switch III to TALK.

(4) Starting at channel 1 position of selector switch IV, perform a two-way speech test for each channel in sequence, observing that each channel gives satisfactory speech quality.

4.8 Vehichlar Transportation test.- The shelter facility shall be secured (and blocked to prevent shifting) to a government furnished 2/12 ton, M-35, truck and transported over the specifically prepared course described in para 4.8.1 for 10 laps at each of the speeds 5 mph, 10 mph, 15 mph and 18-20 mph. A lap is defined as traversing the course in ond direction. The facility shall meet the requirements of paragraph 3.5 after completion of the test. (Note: The shelter facility shall be loaded to simulate the AN/TRC-151(V) using live loads.) (See 6.1)

4.8.1 Vehicular Transportation Course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with 4-inch face fully above ground. The top edges shall have 1 x 1 inch chamfers. The sixth and the twelfth timbers shall be placed at 45 degrees to the travel, and all other timbers shall be long enough to span the vehicle passing over them and shall



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be anchored down securely by any effective means. The surface on which the timbers are placed shall be hard and level.

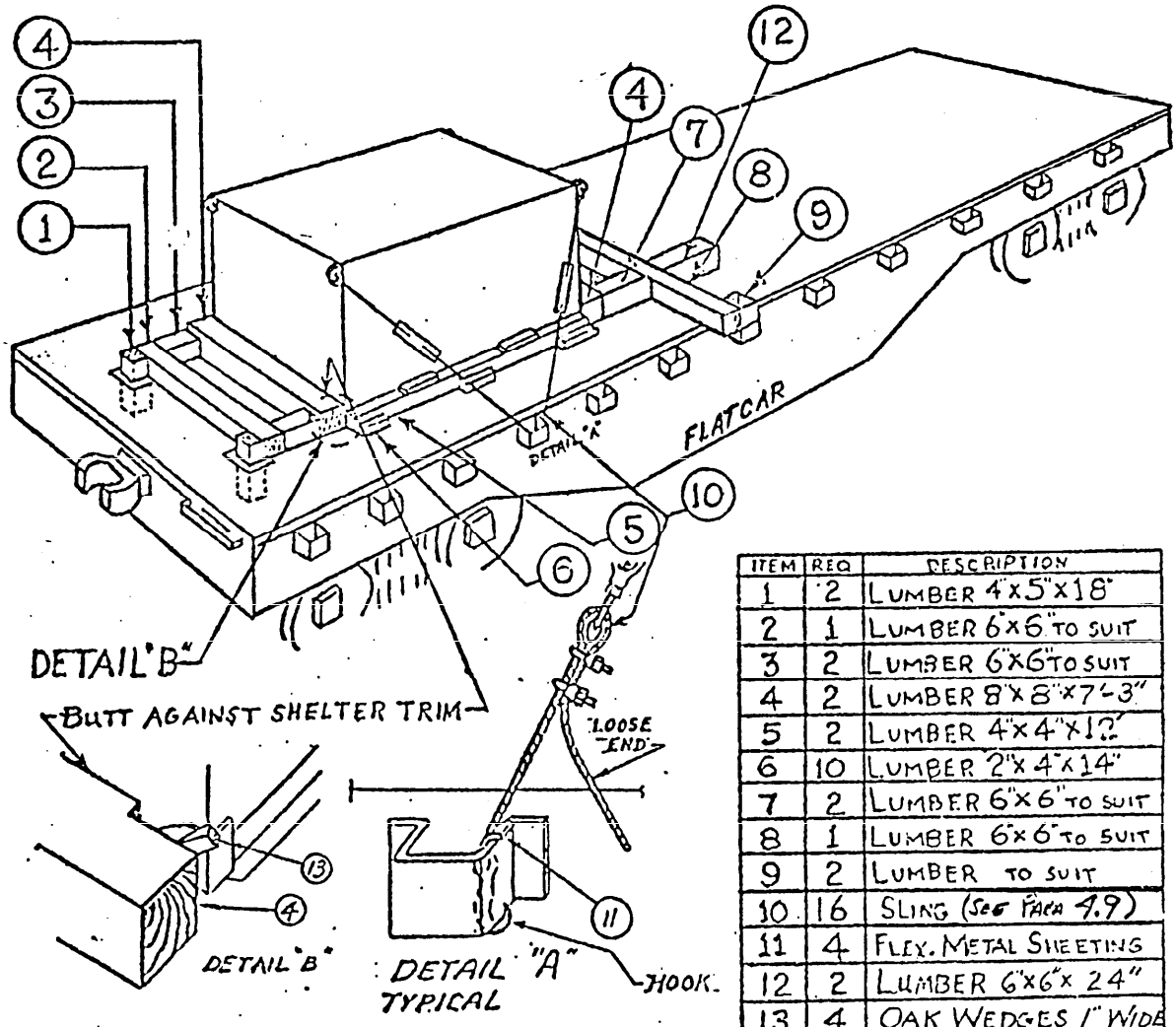
4.9 Rail Transport, Impact test.- The shelter facility loaded as specified in paragraph 4.8 secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tie down slings, shall be as specified in Figure 4 and shall meet the provisions of the test as specified in paragraph 4.9.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the government technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars all with brakes off. The test shall consist of four impacts. The first two impacts shall be with the door end of the shelter facing the loaded gondola car; the second two impacts shall be with the door end of the shelter facing the opposite direction from the loaded gondola car. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.9.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner, along the full length of of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the Figure 4. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckles, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds tension).

4.9.2 Loaded Gondola Car.- The load in the loaded gondola car shall be one of the following:



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| ITEM | REQ | DESCRIPTION |
|---|---|---|
| 1 | 2 | LUMBER 4"x5"x18" |
| 2 | 1 | LUMBER 6"x6" TO SUIT |
| 3 | 2 | LUMBER 6"x6" TO SUIT |
| 4 | 2 | LUMBER 8"x8"x7'-3" |
| 5 | 2 | LUMBER 4"x4"x12" |
| 6 | 10 | LUMBER 2"x4"x14" |
| 7 | 2 | LUMBER 6"x6" TO SUIT |
| 8 | 1 | LUMBER 6"x6" TO SUIT |
| 9 | 2 | LUMBER TO SUIT |
| 10 | 16 | SLING (SEE PARA 4.9) |
| 11 | 4 | FLEX. METAL SHEETING |
| 12 | 2 | LUMBER 6"x6"x 24" |
| 13 | 4 | OAK WEDGES 1" WIDE |

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 13 between towing corner forging and item 4 notch (Detail B)
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 4 SHELTER SECURED ON FLATCAR



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a. A coarse size of either coal, crushed stone, gravel, concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

4.10 Drop tests.-

4.10.1 Flat drop.- The shelter facility loaded as in para 4.8 will be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of para 3.7.

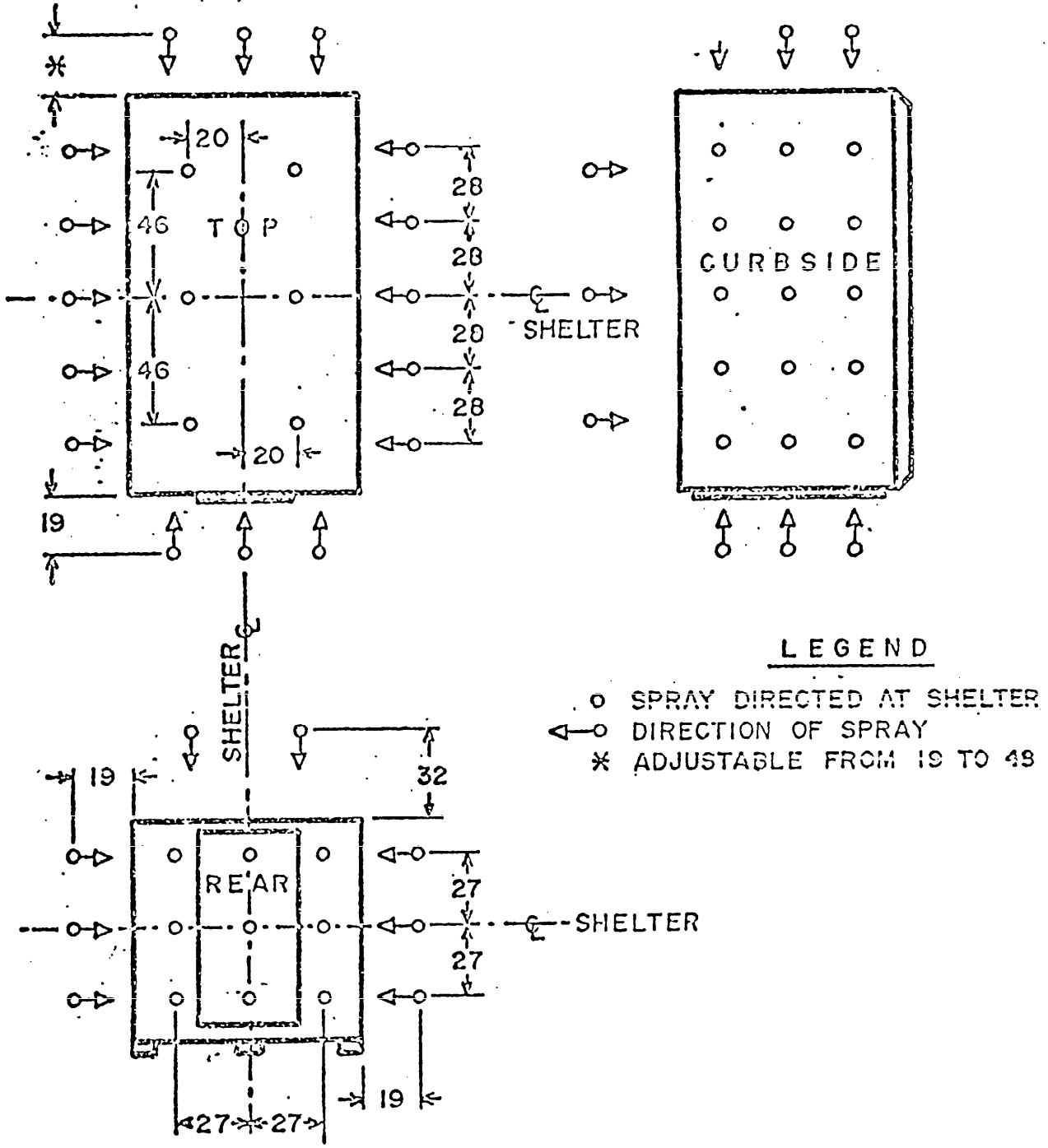
4.10.2 Rotational drop.- The shelter facility loaded as in Para 4.8 shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of para 3.7.

4.11 Watertightness Test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test (NOTE: Perform tests with inside covers of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G 2950 (or GG2950) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 5 for nozzle location Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of para 4.12, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel.) Upon completion of the above test, the shelter facility shall meet the requirements of para 3.8. To determine that no water has entered the shelter panels, holes shall be made at points to be determined



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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE.
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

FIGURE 5 NOZZLE LOCATIONS

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by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the Air Conditioner covers and the door vent panel at the rear of the shelter) and repeat a. above.

4.12 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of paragraph 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tie downs. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.13 Humidity test.- A hardware sample of five of each size and type (if the total quantity of any size or type is less than five, all shall be tested) of bolt, screw, nut, flat washers, lock washers and rivnuts shall be randomly selected for test. The hardware sample shall be subject to the humidity test specified in Method 507, Procedure II of MIL-STD-810 to determine conformance to the requirements of paragraph 3.10.

4.14 Mounting insert tests.-

4.14.1 First article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table VI. After test the requirements of paragraph 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

4.14.2 Group B insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected in each sample equipment. Each selected insert shall be tested in accordance with Table VI. After test the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE VI Mounting Insert Loads

| Insert | First Article and Group C Test | | Group B Test | |
|---|---|---|---|---|
| | Axial (Lbs) | (Torque (inch Lbs) | Axial (Lbs) | Torque (In Lbs) |
| 5/16-18 | 2000 | 100 | 1000 | 100 |
| 1/4-20 | 1300 | 60 | 650 | 60 |
| #10-32 | 600 | 23 | 300 | 23 |



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4.15 Acoustical noise test.- The shelter facility shall be located in an area remote from excessive machinery, vehicle and other noise operating sources to reduce ambient noise to minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band Analyzer, General Radio Corporation, Model 1558A, or equal and Microphone, Piezoelectric, General Radio Corporation, Model 1560-P6 or equal, record ambient noise levels over the frequency ranges specified in 3.12. Operate the exhaust blower & repeat noise level recording at the same frequency levels as above. The noise levels, corrected from ambient shall meet the requirements of 3.12.

4.16 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lug shall be tested for conformance to 3.22.4.

4.17 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.24. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.18 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.25 and Table VII.

TABLE VII - Classification of Visual and Mechanical Defects

| Classification | Defects |
|---|---|
| Major | 1. Mounting plates missing or mislocated.<br>2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.<br>3. Welds - cracked or porous. Inspection of welded assemblies, such as equipment racks, entrance boxes, etc., should be made prior to final paint application.<br>4. Sealer improperly applied (holes, separations, or lack of adhesion).<br>5. Mounting inserts missing or mislocated.<br>6. Burrs or sharp edges in wire duct not removed. |



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TABLE VII - Classification of Visual and Mechanical Defects (cont'd)

| Classification | Defects |
|---|---|
| | 7. Equipments or bracketry loose or missing.<br>8. Doors, etc., inoperative.<br>9. Electrical receptacles, switches, jacks, wire ducts or other electrical components loose, improperly located or inoperative.<br>10. Locking or holding devices missing or inoperative.<br>11. Gaps between sections of wire duct exceed 1/16".<br>12. The blower housing does not align properly with the mounting inserts of the shelter facility.<br>13. Ground stud not properly sealed.<br>14. Bulkhead Power and Signal connector fasteners not sufficiently tight.<br>15. Electrical schematics mission. |
| Minor | 1. Doors not easily operable.<br>2. Fastening devices difficult to operate.<br>3. Finish - abrasions or scratches.<br>4. Burrs and sharp edges on bracket not removed.<br>5. Screws loose.<br>6. Gaskets, not completely and securely adhered.<br>7. Electrical schematics not properly adhered to wall. |

4.19 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.20 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.6 Power and Signal Wiring
Paragraph 4.18 Visual and Mechanical



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5. PACKAGING

5.1 Packaging requirements.- Packaging requirements shall be in accordance with MIL-P-55507.

6. NOTES

6.1 Intended use.- The Shelter, Electrical Equipment S-583( )/TRC-151(V) is the shelter facility portion of the AN/TRC-151(V) and it is used to house the following operational equipments:

a. 2 ea AN/GRC-103 Radio Set

b. 2 ea TD-660( )/G Multiplexer

c. 2 ea TD-754( )/G Multiplexer

d. 2 ea TD-202( )/U or TD-1250( )/U Multiplexer

e. 2 ea CV-1548( )/G Converter, Telephone Signal

f. 1 ea LS-147F/FI Intercommunication Station

g. 1 ea TA-312( )/PT Telephone Set

h. 2 ea TD-1065( )/G High Speed Serial

i. 1 ea TD-1069( )/G Time Division Data Multiplexer (Optional)

j. 2 ea TSEC/KG-27 Security Equipment (Optional)

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads or the government furnished live loads (operational equipment) as specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-583( )/TRC-151(V)

d. A list of equipments to be government loaned for testing purposes.



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6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special test equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.4)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-583R/TRC-151(V). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Tropicalization of material.- All materials of construction are specified on the drawings and are Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus-Resistance satisfactorily in accordance with requirement 508 of MIL-STD-810.

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